

DEVANSHU CHICHOLIKAR

AI ENGINEER | MCP & AGENT SYSTEMS | BOSTON, MA (OPEN TO RELOCATION)

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PROJECTS

- OpenCodeIntel** | *Python, FastAPI, MCP, tree-sitter, Pinecone, Redis, React, TypeScript* | opencodeintel.com **Sep 2024 - Present**
- Shipped a **production MCP server** (12 tools: semantic code search, dependency graph, impact analysis, **context assembly**) enabling AI agents (**Claude, Cursor**) to retrieve codebase context at **p95 208ms**, via a **hybrid AST + embedding RAG** pipeline
 - Benchmarked retrieval to **94% average Hit@1** across **14 OSS codebases** (4 languages, **700-query** eval), improving accuracy **+8.4 points** with cross-encoder **reranking** isolated via a **98-run ablation**
 - Hardened the public **REST API** with dual-mode auth (JWT + SHA-256 keys), two-layer **multi-tenant isolation** (Pinecone metadata filters + Postgres row-level security), and Redis rate limiting; **700+ commits, 450+ tests** solo
- CallBudget** | *Python, FastMCP, scikit-learn, Optuna, Pipecat, Deepgram, DuckDB* **2026**
- Deployed an **agentic system end-to-end** against a real pharmacy-stock workflow: a **FastMCP server** (predict / plan / eval) over a learned stock-probability ranker, cutting expected calls-to-find from **4.3 to 2.3** (about 47%) on a 19-pharmacy simulation
 - Designed a **Claude-driven voice agent** with **guardrails** (calibrated abstention + self-consistency) that cut false-positive-in-stock from about **10% to near 0%**: the reliability layer that makes an agent safe in front of users
 - Hardened the system with an **Optuna**-constrained tuning loop, a held-out validation guard, and a DuckDB observation store (**LLM-as-judge / golden-set eval** discipline)
- tool-crowding** | *Python, MCP, Anthropic API, pytest* | MCP agent-evaluation research **2026**
- Authored a **pre-registered, fail-closed MCP eval harness** (345 tests, 199-tool corpus, 144-trial factorial) measuring agent **tool-selection** under crowding, with **cache-cold enforcement** (per-trial nonce + hard-halt assertion) for byte-level reproducibility
 - Falsified** the naive more-tools-degrade-routing hypothesis with exploratory probes (0 mis-routing at N=6 dissimilar tools), isolating the real failure mode to a task-ambiguity by agent-persona interaction
- SecureScale** | *AWS, Terraform, Packer, Docker, Lambda, RDS, KMS, CloudWatch* **2025**
- Provisioned **multi-AZ AWS infrastructure** (VPC, ALB, ASG, RDS, S3 with KMS) as modular **Terraform** with **IAM least-privilege** security, cutting provisioning from **2 hours to 10 minutes** and **cloud spend 30%** at **99.9% uptime**

SKILLS

AI-Assisted Development: Claude Code (subagents, hooks, custom skills), MCP (Model Context Protocol) server development, **agentic workflows**, spec-first development with AI agents, LLM evals; Cursor

AI & Agents / Eval: agent tool-calling, **RAG** (hybrid AST + embeddings, BM25 + reranking), **LLM evaluation** (calibrated abstention, self-consistency, guardrails, LLM-as-judge), voice agents (Pipecat, Deepgram), tree-sitter, **context engineering**

Languages: Python, TypeScript, Java, JavaScript, Go, SQL

Backend & APIs: FastAPI, Spring Boot, Node.js, REST, gRPC, WebSocket; React, Next.js

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Vercel, Railway

Databases: PostgreSQL, Redis, Pinecone, DuckDB, Supabase, MongoDB

Foundations: Data structures & algorithms, distributed systems, OOP & design patterns, system design

EXPERIENCE

- Northeastern University** **Sep 2025 - May 2026**
Graduate Teaching Assistant, Cloud Computing & Networks *Boston, MA*
- Taught **AWS, Terraform, and distributed systems** to **60+ graduate students**; authored a **Docker + GitHub Actions CI/CD lab** adopted as official course content across 3 sections (**180+ students**)
 - Led **system-design and code reviews** for **15 cloud-native API project teams**, coaching on scalability, fault tolerance, and API design
- Jaksh Enterprise** **Aug 2022 - Jul 2024**
Software Engineer | B2B industrial-equipment platform *Ahmedabad, India*
- Developed a **Java / Spring Boot** product-catalog and quotation engine for **590+ customizable products** serving a **10K+ customer base**, over PostgreSQL and a rules-based pricing module
 - Cut quote-page **p95 latency 65%** (800ms to 280ms) via PostgreSQL indexing, Redis caching, and async processing
 - Owned delivery end-to-end as one of **two contract engineers**: gathered client requirements and shipped via Dockerized **CI/CD**, cutting release cycles from **2 weeks to 3 days**
- Pitney Bowes** **Jan 2022 - Jul 2022**
Software Development Engineer Intern *Pune, India*
- Built **REST APIs** for PitneyShipPro with idempotent endpoints, eliminating **40%** of manual data entry; raised automated **test coverage to 85%** with Jest and Cypress

EDUCATION

- Northeastern University | GPA: 3.85** **Sep 2024 - May 2026**
M.S. in Software Engineering Systems *Boston, MA*
- Coursework:** Generative AI, MLOps, Distributed Systems, Cloud Computing, Database Management Design, Algorithms
- Dhirubhai Ambani Institute of ICT** **Aug 2018 - May 2022**
B.Tech in Information and Communication Technology *Gujarat, India*